NOTE: This worksheet is divided into two parts (Part I and Part II) and it is intended that the first part is done following the slides of the theoretical and practical lectures. In the second part it is intended that the student perform the exercises implementing the subjects addressed and practiced in the first Part.

Part I

Exercise 1

Test the class **`TwoTypePair`** that can be found in the slides 26 and 27 (Lecture 01). Observe the result.

Exercise 2

Test the class **`Pair`** that can be found in the slide 32 (Lecture 01). Observe the result.

Exercise 3

Change the class **`UnorderedPair`** (slide 37, Lecture 01) to make it possible perform two operations:

- Get the value of the first and second element (2 methods).
- Check if the two introduced elements are equal.

Exercise 4

Test the code shown on slide 40 (Lecture 01) to get a message error at compilation time. Can you understand why this error occurs?

Exercise 5

Present a solution to the problem presented in slide 44 (Lecture 01) *"What is the solution that enable* **`drawAll`** *to draw all the shapes?"*.

Part II

Exercise 1

Fill in the blanks:

a) The statement **`Store<T>`** is _____.
b) **`Store`** is _____ and **`T`** is _____.
c) **`Store<String>`** is _____.
d) The use of a parameterized type is known as _____.

Exercise 2

Consider the following fragments of Java code (the first three lines are equal in all fragments, only the last line is different). For each of the fragments, the code compiles correctly? If it compiles, it runs without error or there is any exception?

```
Point[] a = new Point[10];
Object[] b;
b = a;
b[0] = new Point(10,20);
```

```
Point[] a = new Point[10];
Object[] b;
b = a;
b[0] = "Magical Mystery Tour";
```

```
Point[] a = new Point[10];
Object[] b;
b = a;
a[0] = "Magical Mystery Tour";
```

Exercise 3

What happens if we write a code like the presented in the 2nd question but making use of an ArrayList? For example:

```
ArrayList<Point> a = new ArrayList<Point>();
ArrayList<Object> b;
b = a;
b.add(new Point(10,20));
```

Exercise 4

Develop an application that order a list of predefined strings based on their length. Use the `Collections.sort` method. Attention: you should make use of Generics.

Exercise 5

Write a class that act as a library for the following media types: books, videos, and music CDs. Add additional APIs to store and retrieve the media. Attention: you should make use of Generics.